INDIA INNOVATES 2026

Domain:

OPEN INNOVATION (Healthcare)

Team Name:
DIVYA NETRA

MEMBERS & AFFILIATION:

- **Sri Roshan (Team Lead)**
  Space Central School, Sriharikota,A.P
- **Darshika Tharini**
  Space Central School, Sriharikota,A.P
- **Robin Gupta**
  Unitrade Gesellschaft (India) Pvt.Ltd,Delhi

Smart Blind stick

Problem Statement

- In India, around 5 million visually impaired people face daily navigation risks due to traditional sticks.
- Families are always scared that their loved ones will get lost or have an accident.
- Other smart sticks just vibrate or go "beep-beep." This is confusing! The person has to guess what the beep means, which makes them feel anxious and tired.

Existing Gaps:

- Often leads to accidents with unidirectional detection.
- Vibrate when objects are close, but provide no directional guidance or GPS tracking
- Require expensive phones and internet—unaffordable for most Indians
- Currently, there is no easy way to track them in real-time.

Our SOLUTION

- We built our tech into an elbow crutch (like in our prototype photo). This gives much better balance and support than a thin stick.
- We have a small motor that turns our sensor left and right. This allows the crutch to "look around" for a clear path, just like a human neck!.
- It doesn't just beep; it talks! It tells you clearly: "Move Left," "Stop," or "Path Clear." It even works without the internet.
- We use a system. It has a built-in GPS and SIM card. One button sends your exact location to your family instantly without needing a phone.

Honorable Prime Minister of India "Mantras"

- Make In India
- Design For India
- Vocal For Local

Smart Blind Stick: Freedom to move without fear.

Our goal is to help 1 million people walk to school and work independently by the year 2033.

We aren't just selling a piece of metal; we are giving people the freedom to move without fear.

Our Ecosystem

A revolutionary voice-guided, AI-enabled navigation system designed for the unique challenges of the Indian environment.

Active Scanning "Neck"

180° rotating sensor maps the environment to find the safest path, not just obstacles.

Voice Navigation

Offline voice commands like "Turn light", Turn Right, or "Stop" in local languages for zero cognitive load.

One-Touch SOS

Integrated GPS and GSM for real-time tracking and emergency alerts without a smartphone.

Smart Navigation Prototype

Proactive Pathfinding vs. Reactive Beeping

Empowering 1 Million Citizens by 2033

Our architecture

Embedded Systems

C++/Arduino framework for real-time hardware control and sensor fusion.

Edge Computing

On-device logic for obstacle avoidance, ensuring 100% offline reliability.

Geospatial Data

GPS NMEA processing for accurate location tracking and emergency geofencing.

Voice Synthesis

Optimized audio compression for multi-lingual navigation prompts on local storage.

Scalable Tech Stack

Our technology choices prioritize robustness and accessibility, ensuring the device works everywhere in India without internet dependency.

100% Offline Reliability

All processing happens on the edge, removing the "Smartphone involvement" and data dependency.

Technical Specifications

COMPONENT	SPECIFICATION	PRIMARY FUNCTION
Microcontroller	ESP32	Real-time sensor processing & logic
Scanning Module	Servo + Ultrasonic	180° active environment mapping
Connectivity	GSM / GPS A9G Module	Independent SOS & location tracking
Audio Output	DFPlayer + Speaker	Offline voice-guided navigation
Power Source	Li-ion Battery	12+ hours of active operation

Core Processing
Low-latency sensor fusion and power management.

GPS/GSM Integration
Independent cellular connectivity for emergency SOS.

Offline Reliability

It is engineered for the "Real India," where internet connectivity is often unreliable. Our hardware-centric approach ensures 100% functionality in any environment.

Edge Computing

All obstacle avoidance and pathfinding logic are processed locally on the device, ensuring zero-latency response.

Independent Safety

Built-in SIM module removes the "Smartphone," making safety features accessible to all socio-economic groups.

40%

Reduction in Navigation Errors

Unique Value Proposition

We stand out by offering a unique combination of physical stability and digital intelligence.

360° Awareness

The rotating sensor provides a "radar-like" view, detecting obstacles that static sensors miss.

Zero Learning Curve

Natural language voice guidance makes the device usable by everyone within minutes.

Phone-Free Independence

The built-in cellular module removes the "Smartphone involvement," making safety features accessible.

Prototype Testing Video -1

Click the Link

https://drive.google.com/file/d/1d8cNFftsOTIIeANE9hWHmWSmSWFBaDuz/view

Prototype Testing Video -2

Click the Link

https://drive.google.com/file/d/1cKk-m7I5UbL9m3ADw6F8MvHAELBlpMZ1/view

Market Opportunity & Financials

TAM: Total Visually Impaired (India)	5 Million
SAM: Active Walkers (Aged 10-65)	3.2 Million
SOM: Target (First 3 Years)	5,00,000 Units

Metric	Retail Model	B2G Model
Manufacturing Cost	₹3,314	₹3,314
Selling Price	₹4,500	₹4,143
Profit Margin	35.8%	25.0%

Distribution Strategy

B2G: Government Welfare

Leveraging the ADIP Scheme and State Welfare Departments for bulk procurement and distribution.

CSR & NGO Partnerships

Collaborating with corporate giants (Ex: Tata, ONGC) to reach rural communities via CSR initiatives.

Direct-to-User

Targeting urban professionals and students through specialized assistive technology stores.

₹25L
PRE-SEED ASK

1,250
UNITS TO BREAK-EVEN

"Not seeing doesn't have to mean staying at home."

Join us in making India's streets accessible for everyone.

THANK YOU

Any Questions?

REFERENCES / LINKS

DATAS LINK
https://www.pib.gov.in/PressNoteDetails.aspx?NoteId=156810&ModuleId=3&reg=1&lang=9
PRESENTATION LINK p
https://docs.google.com/presentation/d/1h2pqAwBG1HiqV27X0LYhA8EpLRhNy1qe/edit?usp=sharing&ouid=112006228167084072552&rtpof=true&sd=true